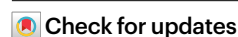


From least-cost to SDG-optimal sectoral allocation of Paris Agreement-compatible mitigation efforts

Received: 26 August 2025

Accepted: 25 February 2026

Published online: 07 April 2026



Dirk-Jan Van de Ven¹✉, Clàudia Rodés-Bachs¹, Théo Rouhette¹, Russell Horowitz¹, Jon Sampedro^{1,2}, Alexandros Nikas³, Natasha Frilingou³, Xin Zhao⁴, Abhishek Chaudhary⁵, Gokul Iyer⁴, Jorge Moreno¹ & Konstantinos Koasidis³

Limiting global warming to well below 2 °C necessitates profound decarbonization, but how to distribute mitigation efforts over sectors remains a widely debated issue. Although integrated assessment models traditionally rely on ‘least-cost’ optimization to answer this question, the resulting sectoral allocations vary widely and ignore impacts on other potential policy objectives. Here we connect an integrated assessment models with a portfolio analysis to evaluate how sector-specific mitigation actions impact key indicators from Sustainable Development Goals (SDGs) related to poverty, health, water, economy and land, and to identify Pareto-optimal and Paris-compliant mitigation portfolios that reveal the trade-offs between other sustainable development priorities. Furthermore, we define ‘SDG-balanced’ portfolios that, in most cases, outperform standard least-cost scenarios across all five SDG indicators for an equivalent carbon budget. Our findings demonstrate that the simultaneous evaluation of a broader set of policy priorities is crucial to provide truly policy-relevant guidance for the climate transition.

In 2015, governments acknowledged the urgency to foster human well-being within planetary boundaries by adopting the Paris Agreement’s goal to limit global temperature increase to well below 2 °C, while pursuing efforts to stay below 1.5 °C (ref. 1) and the 2030 Agenda for Sustainable Development, comprising 17 Sustainable Development Goals (SDGs), promoting sustainable growth, social well-being and environmental protection². Several SDGs have been shown to interact with climate change^{3,4} and action (SDG13)^{5–11}. These include, but are not limited to, indicators related to land (SDG2/food and SDG15/terrestrial biodiversity), water (SDG 6/water), health (SDG 3/air pollution-driven mortality) and the economy (SDG 8/economic growth), with a range of potential interactions—both trade-offs and synergies.

Although achieving Paris Agreement-compatible mitigation levels requires deep transformations in virtually every sector, the widely debated issue of how to allocate contributions across economic sectors remains open. The contributions of energy supply, agriculture, forestry and other land uses (AFOLU) and the three main energy end-use sectors (industry, buildings and transport) differ widely between existing mitigation scenarios^{12,13}. Indicatively, the energy supply and AFOLU sectors provide an important negative emissions potential to offset emissions in end-use sectors¹⁴. Depending on the negative emission technologies (NETs) applied—for example, bioenergy combined with carbon capture and storage (BECCS), large-scale afforestation or direct air capture (DAC)—this potential can feature risks and trade-offs, such

¹Basque Centre for Climate Change, Leioa, Spain. ²IKERBASQUE, Basque Foundation for Science, Bilbao, Spain. ³National Technical University of Athens, Athens, Greece. ⁴Center for Global Sustainability, University of Maryland, College Park, MD, USA. ⁵School of Management and Law, Zurich University of Applied Sciences, Winterthur, Switzerland. ✉e-mail: dj.vandeven@bc3research.org

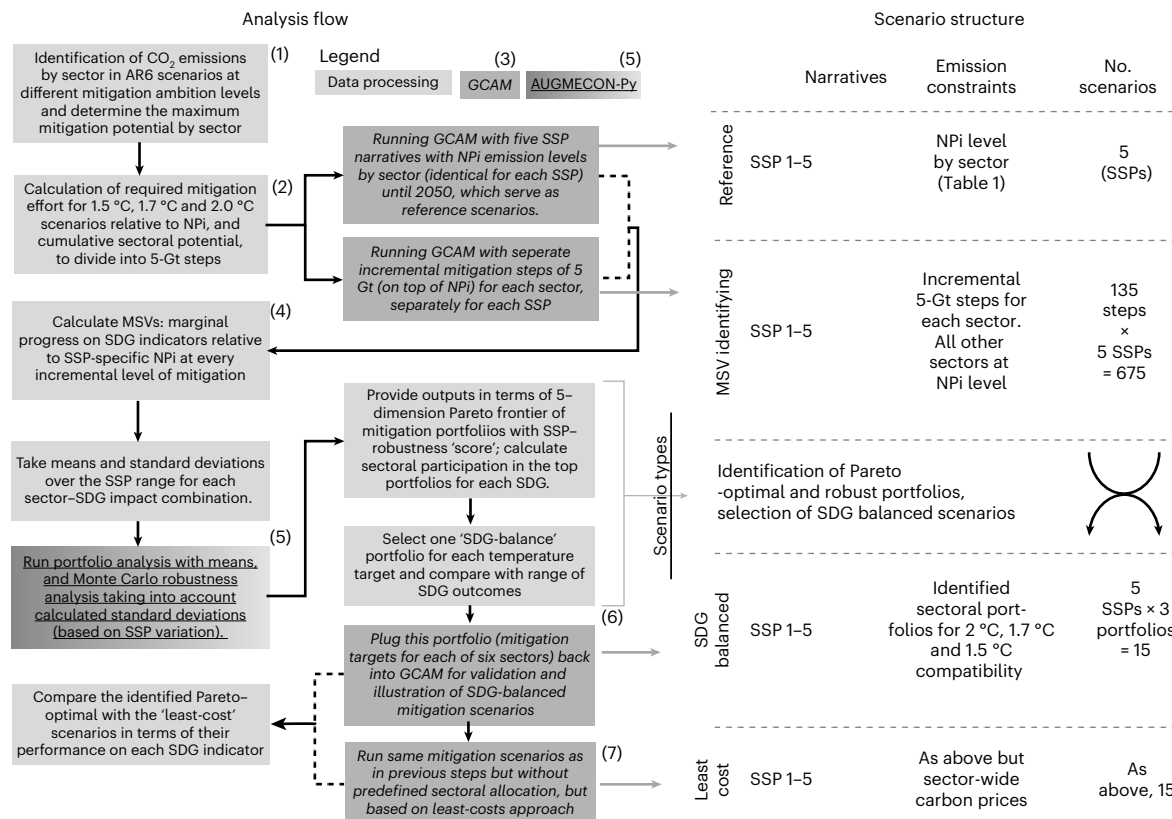


Fig. 1 | Analysis flow and scenario structure. The numbered steps refer to their respective section in the Methods: (1) ‘Sectoral mitigation potential’, (2) ‘Incremental scenarios and emission budgets’, (3) ‘IAM environment’, (4) ‘SDG

indicator choice and definition’, (5) ‘Robust portfolio analysis’, (6) ‘Selection and model assessment of an ‘SDG-balanced’ portfolio’ and (7) ‘Defining and running comparable mitigation scenarios based on least-cost allocation’.

as water stress, biodiversity loss or increased energy demand^{15–18}. As such, different sectoral allocations of mitigation efforts may have markedly different impacts on economic costs, technologies employed and non-climate outcomes (for example, food production, health and employment). A quantitative assessment of co-benefits and trade-offs driven by different mitigation efforts in core economic sectors can, therefore, offer valuable insights into their contributions to future mitigation strategies.

Detailed-process integrated assessment models (IAMs) can play an important role in designing effective global mitigation scenarios. By covering all energy-consuming, greenhouse gas (GHG)-emitting sectors and featuring sufficient technological detail, IAMs are well suited to identify the sectoral mix of economy-wide mitigation efforts^{19,20}. Despite employing markedly different economic theories²¹ or internal logic and structures²², many IAMs apply some form of least-cost approach for defining where each additional unit of mitigation takes place^{23,24}. However, even assuming IAM-derived technology cost estimates are of reasonable quality to provide meaningful least-cost mitigation pathways—a vividly debated assumption in terms of reliability²⁵ and timely relevance¹³—a least-cost option may not always be optimal in terms of other policy objectives, such as those relating to climate-interacting SDGs, such as health and well-being²⁶, poverty eradication²⁷, biodiversity protection²⁸ or tackling water scarcity²⁹.

Here, we explore how mitigation efforts towards 1.5 °C, 1.7 °C and 2 °C-compatible futures can be allocated across sectors to simultaneously advance broader SDGs (Fig. 1). Starting from a pathway based on current national policies and short-term pledges—which remain largely insufficient^{30,31}—we quantify both the required sectoral emission shares and the maximum mitigation potential per sector. Beyond post-scenario assessments of SDG impacts as typically pursued^{3,11}, our

analysis applies the concept of ‘marginal SDG emissions-reduction values’ (MSVs)⁷ to estimate the marginal impact of sector-specific CO₂ mitigation on five non-climate SDG-relevant metrics until 2050: food and home energy expenditures as a share of income (SDG 1), premature mortality from ambient air pollution (SDG 3), physical water scarcity (SDG 6), gross domestic product (GDP) per capita growth (SDG 8) and potential species loss (PSL) from land-use change (LUC) (SDG 15) (see ‘SDG indicator choice and definition’ section in the Methods for rationale and details), separately for five Shared Socioeconomic Pathways (SSPs) to additionally evaluate the robustness of such impacts. We then construct optimal and robust sectoral mitigation portfolios adapting an established approach combining the Global Change Analysis Model (GCAM) and multi-objective optimization theory³² to simultaneously achieve climate targets and advance progress on non-climate SDGs³³. We identify ‘SDG-balanced’ portfolios that perform reasonably well over all SDG-related indicators, which are then reassessed in GCAM for methodological validation and comparison with model-assessed least-cost mitigation pathways towards the same climate targets.

Taking stock of mitigation potentials

To keep our analysis within feasible limits, we first identified the sectoral mitigation potentials over 2020–2050, relative to sectoral emissions, in a National Policies (NPI) scenario, as well as the total additional mitigation efforts required to 2050 (Table 1 and Extended Data Fig. 1; ‘Sectoral mitigation potential’ section in the Methods). This sectoral mitigation potential stock-taking exercise shows that the potential mitigation contributions differ strongly among sectors, with energy supply (including energy extraction, electricity and refining) having the greatest potential (~270 GtCO₂), while buildings and DAC can only contribute around 35–40 GtCO₂ if applied at their assessed maximum

Table 1 | The means of emissions (GtCO₂) by sector and scenario, and the calculation of the sectoral mitigation potentials (GtCO₂) and total budget (GtCO₂), including the required mitigation relative to NPi in each temperature goal

Sector	Scenario	2019–2021 observation value (EDGAR)	Mean pathways			Total 2020–2050 budget use	Mitigation with respect to NPI ^c (5-Gt rounding)
			2030	2050	(No. of observations)		
GtCO ₂							
Industry	NPI ^e	9.6	11.7	13.0	(18)		
	2°C pathways		9.2	6.8	(20)		
	1.7°C pathways		7.7	4.2	(25)		
	1.5°C pathways		7.0	2.5	(28)		
	Minimum benchmark ^a		7.2	1.2			180
Buildings	NPI ^e	3.4	3.3	3.6	(16)		
	2°C pathways		3.0	2.5	(23)		
	1.7°C pathways		2.9	1.9	(28)		
	1.5°C pathways		2.5	1.2	(29)		
	Minimum benchmark ^a		2.6	0.6			40
Transportation	NPI ^e	7.9	8.2	7.8	(18)		
	2°C pathways		8.2	7.1	(33)		
	1.7°C pathways		7.7	5.4	(37)		
	1.5°C pathways		7.2	3.9	(37)		
	Minimum benchmark ^a		7.0	1.9			65
AFOLU	NPI ^e	3.5	2.5	0.4	(17)		
	2°C pathways		1.5	−0.3	(38)		
	1.7°C pathways		0.8	−0.5	(44)		
	1.5°C pathways		0.1	−2.0	(43)		
	Minimum benchmark ^a		−0.4	−3.8			85
Energy supply	NPI ^e	15.4	14.4	14.2	(18)		
	2°C pathways		11.9	2.8	(36)		
	1.7°C pathways		11.1	1.2	(42)		
	1.5°C pathways		9.0	−1.3	(39)		
	Minimum benchmark ^a		9.0	−4.6			270
DAC ^d	Minimum benchmark ^a		−0.1	−5.0			35
Sum of all sectors	Minimum benchmark ^a						675
Total for mitigation targets	NPI	40.7	39.8	38.1		1,196.1	
	2°C pathways		33.7	18.8		897.1	300
	1.7°C pathways		30.2	12.3		779.8	415
	1.5°C pathways		25.9	4.4		636.1	560

Means for NPi, 2°C, 1.7°C and 1.5°C pathways are taken from the AR6 scenario database⁴². The observation count of every mean is reported in parentheses. ^aThe Minimum benchmark is calculated as the lower 10% point of sectoral emissions in 2050 in the range of 1.5°C-compatible scenarios and is used to determine the sectoral potential. The 2030 value in this benchmark is calculated adapting the path to the 2050 level departing from observed 2020 emissions, hence it can be above the 2030 value for 1.5°C pathways. See the Methods for details. ^bThe total budget use is calculated assuming linear interpolation between the observed 2020 value and 2030 totals and between the 2030 and 2050 totals. ^cThis column only mentions the values relevant for this study. See the Methods for the translation of sectoral pathways to cumulative-emission potentials. ^dDAC is excluded from the AR6 pathway assessment owing to few models reporting values. The Minimum benchmark is based on predefined future mitigation potential by Fuss et al.⁴³. ^eSee Extended Data Fig. 2a for a graphical representation of mitigation in the NPi scenario.

potential (see Table 1 for a complete overview). Also, energy supply and AFOLU feature the largest negative emissions potentials, contrary to buildings and transport. In particular, limiting the mean global temperature increase to 2°C, 1.7°C or 1.5°C would require significant additional mitigation efforts beyond what NPi would achieve (Extended Data Fig. 2) by 300, 415 or 560 GtCO₂, respectively.

Marginal SDG emissions-reduction values per sector

We then evaluated the marginal SDG impacts at incremental sectoral mitigation efforts (in 5 GtCO₂ steps). These MSVs, reflecting the

marginal impact on an additional unit of mitigation on various other SDG progress indicators⁷, are identified for five SDG indicators, six sectors and all five SSPs³⁴, with the latter providing a range for many important context variables, such as demand-side changes, pollution controls and agricultural yields, feeding through in the relative performance of sectoral mitigation for selected SDGs (Fig. 2). Within each sector, a wide mix of mitigation options is deployed to achieve mitigation targets, while the precise mix, depending upon the ambition level and SSP narrative (see Extended Data Fig. 3 for a graphical example for the transportation sector), determines the variability in SDG outcomes.

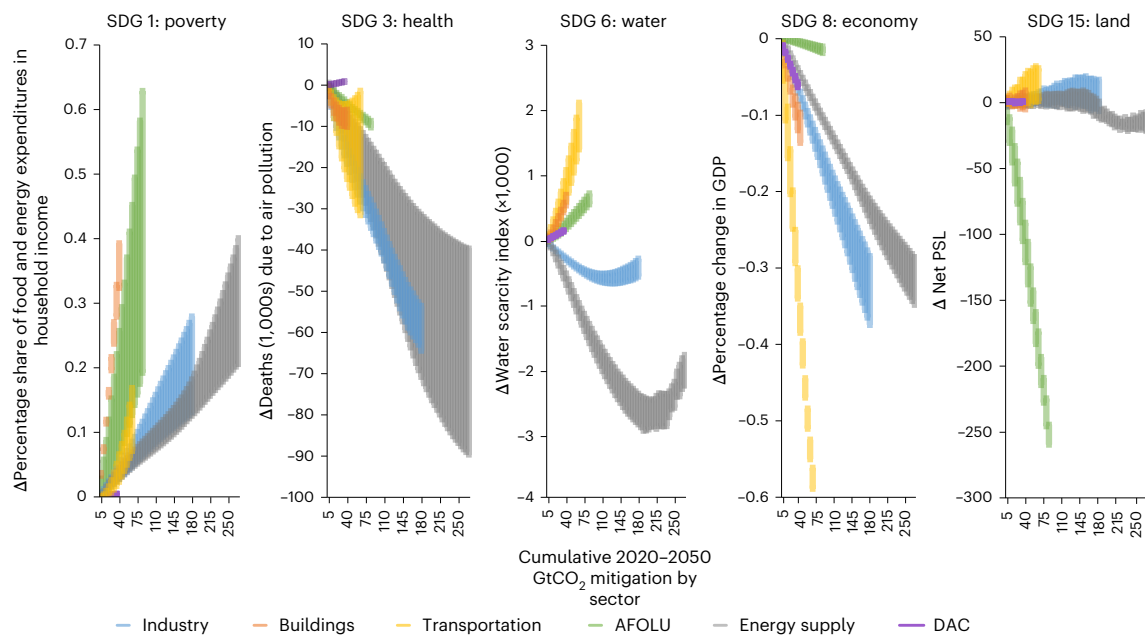


Fig. 2 | Marginal impacts in five SDG indicators from incremental mitigation efforts in each sector (MSVs). The ranges represent the range of outcomes under five different SSPs relative to their respective NPi reference scenario (Fig. 1 and Extended Data Fig. 2). The estimates are calculated by applying incremental steps

of mitigation for each of the six sectors in GCAM and measuring the incremental outcomes per SDG ('Incremental scenarios and emission budgets', 'IAM environment' and 'SDG indicator choice and definition' sections in the Methods).

The results indicate considerable trade-offs on SDG performance driven by mitigation in different sectors. For example, transport mitigation can be most effective towards better health by phasing out internal combustion engine vehicles. However, mitigation in transportation has negative trade-offs with the selected economy- and water-related SDG indicators owing to high investment requirements and demand for water-intensive biofuels, respectively. Mitigation in AFOLU, however, largely achieved by land conservation in the applied modelling context, has significant biodiversity co-benefits but does not significantly affect GDP nor bring about health benefits, while greatly fuelling poverty through higher food prices. Likewise, climate change mitigation in the built environment also exacerbates poverty due to higher implied costs for home energy, while performing poorly in water access and economic growth. Mitigation in energy supply and industry—that is, the sectors traditionally of focus in mitigation efforts since the Kyoto Protocol³⁵—is initially beneficial for the selected health and water objectives by reducing reliance on thermal fossil technologies (until bioenergy comes in at scale), without threatening poverty, GDP and biodiversity much. Finally, compensating emissions through DAC comes with moderate impact on GDP—partly because it delays mitigation towards the 2040s—and hardly affects poverty. However, expectedly, it yields no co-benefits for the selected health, water or biodiversity indicators.

Pareto-optimal mitigation frontiers

To balance these trade-offs and leverage co-benefits, we applied a multi-objective optimization analysis with these MSVs and find five-dimensional (one dimension per SDG indicator) Pareto frontiers (that is, sets of Pareto-optimal solutions outlining the full spectrum of what can be meaningfully achieved in each dimension, without sacrificing the performance of others) of the 2 °C-, 1.7 °C- and 1.5 °C-compliant mitigation portfolios (Fig. 3a). A robustness analysis was additionally applied via Monte Carlo simulations, by considering socioeconomic uncertainty in the MSVs, as reflected in the different SSPs, in turn indicating certain mitigation portfolios as more robust to SSP variability.

Our results indicate that, relative to an NPi scenario, a 1.5 °C-compliant scenario would annually reduce premature mortalities by

120,000–150,000 avoided deaths, while reducing PSL by 200–260 species on average, in 2020–2050 (the ranges indicate the variety in sectoral mitigation mixes). However, such a scenario would also lead to increased home energy and food expenditures relative to income (0.7–1.3% global average, implying much higher percentages for those already in poverty), while yielding a negative impact on global GDP, with an average reduction of 0.8–1.3% over that period. The impacts on water scarcity, as expressed by annual water withdrawals divided by the total renewable water supply, range from a deterioration of up to -1.98×10^{-3} to an improvement of 0.76×10^{-3} . By relaxing the temperature constraint to 1.7/2 °C of warming, we observe similar trends for socioeconomic and environmental indicators, albeit to a lesser extent, that is, lower environmental gains in conjunction with lower economic losses.

We should note that the best and worst performances of sustainability indicators cannot be simultaneously achieved—for example, the least amount of economic deterioration cannot necessarily be combined with the best performance on avoided mortalities (at least not for all indicators at the same time)—with the dynamics and trade-offs among the five dimensions showcasing considerable complexity, as also highlighted in the Pareto-type plots. The curved shape of the 1.5 °C Pareto frontier relative to the other two targets (see the empty space in the region with the lowest negative economic and poverty impacts), indicates that the observed trade-offs are markedly harder to bridge, highlighting profound challenges associated with pursuing the 1.5 °C target (Fig. 3). This is also reflected in the number of Pareto-optimal solutions calculated, with the 1.5 °C case comprising markedly fewer solutions (3.9 million as opposed, for example, to 6 million in the 1.7 °C case). As expected, this stresses that the number of potential sectoral mitigation portfolios diminished with the stringency of the temperature target (also considering the large number of iterations performed).

In terms of robustness, measured by the number of iterations in which a certain portfolio is Pareto optimal, considering cross-SSP uncertainty, we observe that scenarios favouring certain SDG dimensions are more robust. For example, biodiversity (measured in PSL) de facto improves with stronger mitigation in AFOLU. Hence,

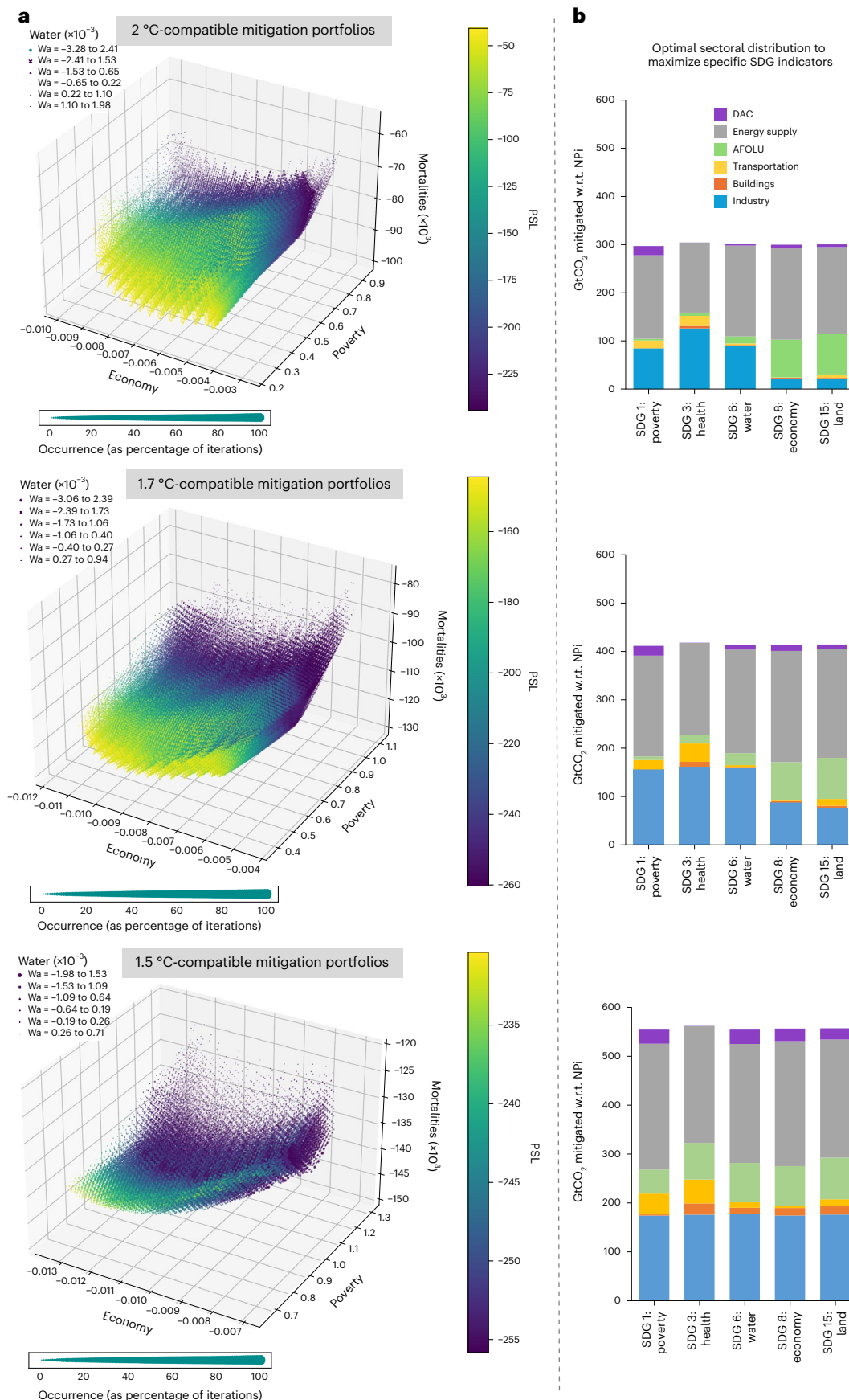


Fig. 3 | Mitigation portfolios at the SDG-optimizing Pareto frontier. a, The Pareto frontiers over five different SDG dimensions for mitigation efforts compatible with three temperature targets. The axis colours retain the same meaning for all three targets although the axes are rescaled. Occurrence,

measured by the size of the bubbles, determines the robustness of each portfolio. **b,** For each temperature reduction target and SDG, the sectoral contribution of the top 10% best-performing portfolios (weighted by occurrence) on that SDG are shown. w.r.t., with respect to.

competition between sectoral portfolios maximizing performance over biodiversity is lower and, consequently, robustness of such portfolios is higher. By contrast, in the case of premature mortality, there is significantly stronger conflict between sectors and hence the robustness of portfolios prioritizing health is overall lower.

SDG-defined mitigation portfolios

The top 10% performing portfolios on each SDG, weighted by portfolio robustness, highlight sectoral distributions when performance is clearly prioritized for one specific SDG (Fig. 3b). We find variability in sectoral distribution decreases with more stringent mitigation objectives, reflecting the requirement for deep decarbonization that essentially exploits almost all mitigation potential (Table 1) in virtually every sector to limit the global temperature increase to 1.5 °C.

Overall, featuring the largest mitigation potential, energy supply has the highest contribution across portfolios—over half of total effort in most portfolios delivering 2.0 °C- and 1.7 °C-compatible pathways, and around 250 GtCO₂ in all 1.5 °C-compatible portfolios. Interestingly, this 250 GtCO₂ point is where the energy supply sector enters the net-negative emissions regime with high BECCS levels, and SDG performance begins to worsen in most dimensions (Fig. 2). Industrial mitigation is nearly maxed out in 1.5 °C-compatible pathways irrespective of prioritized dimensions. For portfolios prioritizing poverty-, health- and water-related SDGs, industry is also relatively high in the 2.0 °C- and 1.7 °C-compatible pathways. However, for portfolios prioritizing economy- and biodiversity-related SDGs, it is significantly lower in the 2.0 °C- and 1.7 °C-compatible pathways. This indicates that the prominent role in 1.5 °C-compatible pathways is largely due to exhausted mitigation potential in the AFOLU sector, which performs best on these SDGs. Similarly, the exhausted mitigation potential in industry leads to high AFOLU mitigation in 1.5 °C-compatible pathways prioritizing poverty, health and water goals. The building sector only has a relevant contribution to mitigation when health is prioritized, while transport is contributing to portfolios benefiting poverty eradication and health. Finally, the role of DAC clearly increases with deeper mitigation: it shows a relevant contribution in 2 °C-compatible portfolios focusing on poverty but depletes its maximum potential in 1.5 °C portfolios, prioritizing every dimension except health.

Three sectors have the potential to contribute to negative emissions: AFOLU through, for example, afforestation, energy supply through BECCS and DAC. We observe that the role of these sectors, and hence the focus on NETs, is largest in portfolios prioritizing economic growth and biodiversity protection, while the role of energy end-use sectors, which do not largely rely on NETs, is clearly largest when prioritizing health. For 1.5 °C-compatible scenarios, the potential of NETs is nearly maxed out when focusing on any dimension except health (since NETs typically only capture CO₂ without reducing other pollutant concentrations).

SDG-balanced and comparable least-cost mitigation portfolios

For illustrative and validation purposes, an ‘SDG-balanced’ mitigation portfolio was selected for each temperature target that displays the best weighted performance across all indicators, relative to the best and worst performing portfolio for that SDG (Extended Data Table 1; ‘Selection and model assessment of an ‘SDG-balanced’ portfolio’ section in the Methods). Our portfolio approach assumes sectoral mitigation increments as linearly additive, ignoring potential interactions between sectors feeding through in SDG performance. Therefore, by simulating these complete SDG-balanced mitigation portfolios in GCAM, allowing for sectoral interactions, we validate the results on SDG impacts from the separate sectoral mitigation runs. For comparison, we also simulate the same total CO₂ emission constraints as in these SDG-balanced scenarios in a traditional ‘least-cost’ mitigation setting, with a single sector-wide carbon price (‘Defining and

running comparable mitigation scenarios based on least-cost allocation (step 7)’ section in the Methods).

Most mitigation efforts in the SDG-balanced portfolios, in addition to the effort assumed in NPi (Extended Data Fig. 2), would go towards energy supply, industry and AFOLU; only the 1.5 °C-compatible portfolio reflects additional efforts in buildings, transportation and DAC (Fig. 4a). Instead, least-cost scenarios divide additional mitigation efforts more equally, giving more weight to mitigation in the built environment, transportation and through DAC, at the expense of mitigation in energy supply (2 °C-compatible scenario) or industry (1.7 °C- and 1.5 °C-compatible scenarios). The key distinguishing factor between SDG-balanced and least-cost scenarios is that the former can differentiate carbon prices across sectors (Fig. 4b). This flexibility implies that, to optimize over SDG outcomes, a relatively lower carbon price is charged to the building and transportation sectors. The weighted-average carbon price across sectors is higher than the sector-wide price in least-cost scenarios for 1.7 °C- and 1.5 °C-compatible scenarios, as expected since the logic of least-cost scenarios is that every subsequent unit of emission abatement occurs where this is most cost-effective. Nevertheless, for the selected 2 °C-compatible scenario, the SDG-balanced weighted-average carbon price is slightly below the least-cost carbon price, driven by high disparities between sectors in their responsiveness to relatively low carbon prices.

SDG-balanced portfolios perform relatively well across all five SDG indicators compared with the maximum and minimum performance of all mitigation portfolios on these indicators (identified in the calculated Pareto frontiers) (Fig. 4c). Relative to the impacts of least-cost scenarios, SDG-balanced scenarios perform either slightly or significantly better on all SDG indicators, except only poverty in the 1.5 °C-compatible scenario. This confirms that our methodology—that is, selecting mitigation portfolios based on relative performance towards SDG targets—indeed allows identifying more effective mitigation portfolios in terms of the selected SDG indicators. Overall, for some indicators (generally those improving with mitigation such as health and biodiversity), we observe that deeper mitigation brings the performance closer to the maximum for that objective, while for other indicators such as economic growth and poverty eradication the opposite is true. This pattern essentially shows that, at lower mitigation ambition, better health and biodiversity outcomes may be feasible but at the cost of lower performance on other SDGs, while this balance of trade-offs changes with deeper mitigation.

Finally, we find that, overall, the model simulation of the SDG-balanced portfolio closely mimics the calculated impact based on the sum of separate sectoral impacts, relative to the overall range of results (Fig. 4c). This validates our methodology of choosing portfolios based on independent sector-specific MSVs, implicitly assuming them as linearly additive. For some objectives, sectoral interactions amplify the SDG impact, either positively (health and biodiversity) or negatively (poverty eradication). An example for the amplifying effect is that simultaneous mitigation in several sectors would require more land conservation to compensate for LUC emissions from bioenergy production.

This simulated SDG-balanced portfolio also allows to illustrate what such a scenario—found to outperform a traditional least-cost scenario with the same emission budget—would entail in terms of emissions by sector (Fig. 5a) and outcomes of SDG indicators (Fig. 5b) over time. Emissions drop to -17 GtCO₂, -10 GtCO₂ and net-zero in 2050 for respectively 2 °C-, 1.7 °C- and 1.5 °C-compatible SDG-balanced portfolios, respectively. The net-positive emissions in end-use sectors (buildings, transportation and industry) are entirely compensated by net removals in energy supply, AFOLU and DAC in the 1.5 °C-compatible portfolio, while only partially by AFOLU in the less ambitious portfolios. Poverty-, water- and economic growth-related objectives are, over time, increasingly affected by mitigation, albeit moderately in absolute terms. Health and biodiversity objectives increasingly benefit from

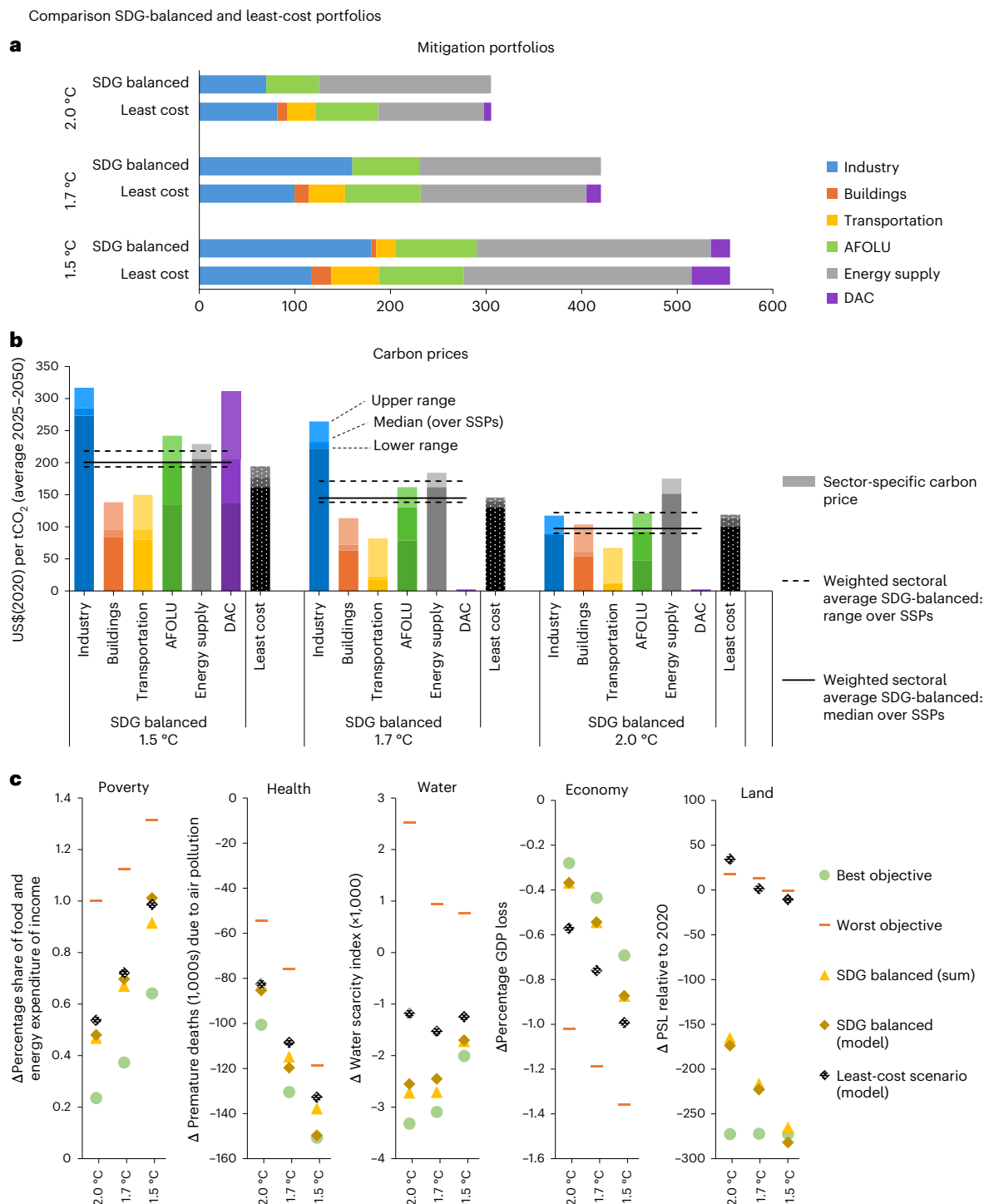


Fig. 4 | Comparison of SDG-balanced mitigation portfolios and least-cost portfolios achieving the same overall emissions target. a, The sectoral distribution of mitigation relative to the NPI scenario (Extended Data Fig. 2), for three temperature goals. **b,** The (sectoral) carbon prices (averaged over 2025–2050) required to achieve the (sectoral) mitigation targets, including an average price weighted by the relative sectoral share of ‘mitigable’ emissions in 2020. Mitigable emissions are either the total emissions in 2020 for sectors that do not achieve negative emissions or the absolute difference between 2020

emissions and 2050 emissions if the latter is negative. The range reflects the full min–max range. **c,** The performance of the SDG-balanced mitigation portfolios relative to the full range of outcomes (from ‘best’ to ‘worse’) from portfolios at the Pareto frontier for every objective. For SDG-balanced portfolios, the ‘sum’ outcome based on the sum of the MSVs is shown (Fig. 2), and the ‘model’ outcome having reiterated the entire mitigation portfolios back into the GCAM model. The modelled outcome is also included for least-cost scenarios. See the Methods for the selection of the of the SDG-balanced scenario.

mitigation efforts. The uncertainty in background parameters, measured through the different SSP narratives, logically also increases over time and is the largest for poverty eradication and health objectives, while relatively small for economic impacts.

Discussion

Climate change mitigation pathways quantified in IAMs traditionally allocate mitigation efforts among sectors based on an economy-wide cost-minimizing principle. While this ability to find least-cost pathways

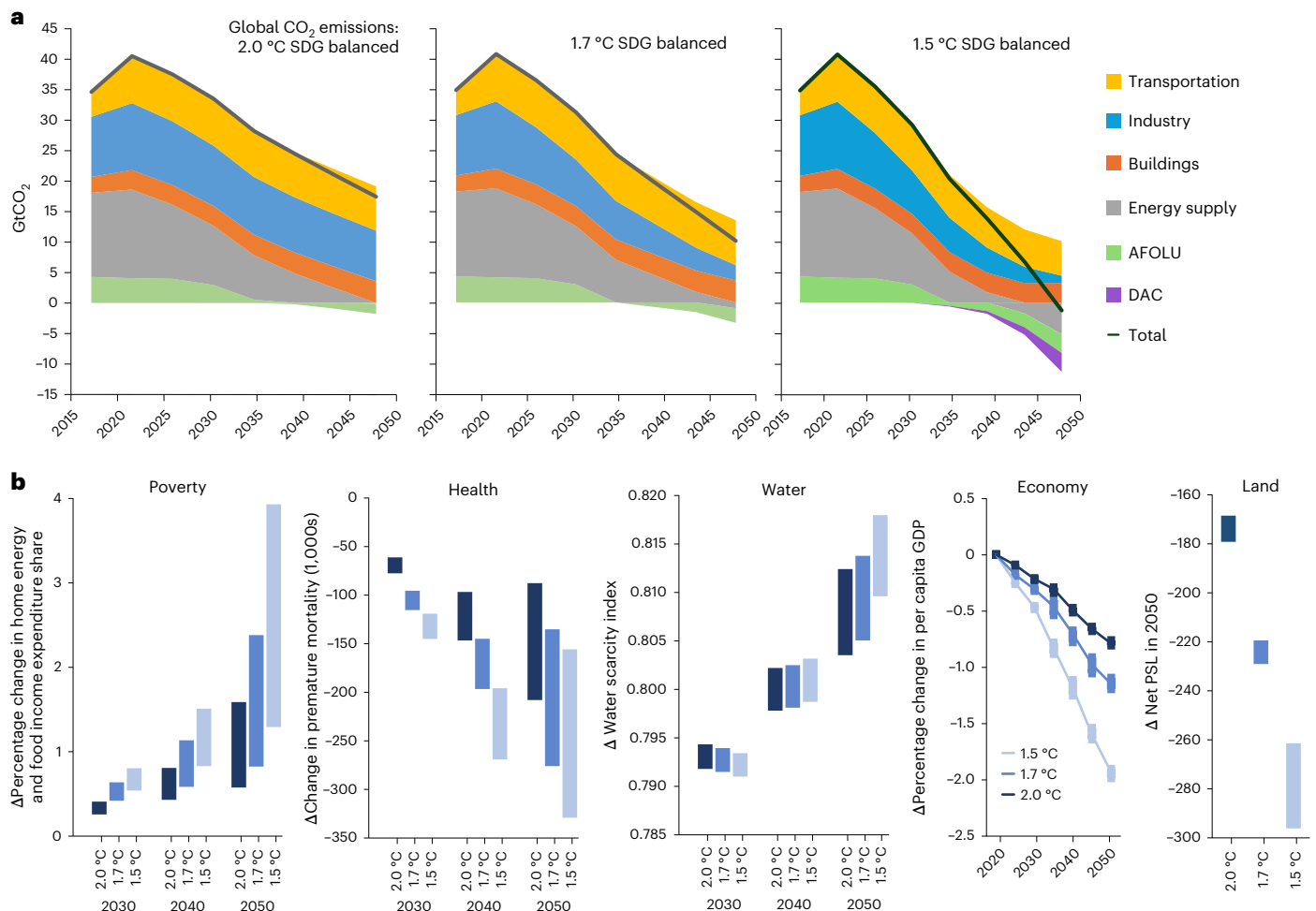


Fig. 5 | Illustration of scenarios applying 2 °C, 1.7 °C and 1.5 °C-compatible SDG-balanced mitigation portfolios on top of NPi mitigation. a, CO₂ emissions by sector over time, which are an identical input for all SSP runs. **b**, The performance on SDG indicators over time relative to the SSP-specific NPi scenario, reflected by a range of outcomes covering all five SSPs.

is a pivotal quality of such models, cost-optimal pathways may not necessarily be optimal from the viewpoint of other policy priorities. Despite growing literature on the SDG-related implications of IAM-based mitigation pathways, SDG performance is typically assessed *ex post*—that is, after obtaining the least-cost mitigation pathway and extracting its model variables relating to non-climate objectives^{5–9}. While some exercises have evaluated SDG performance over a range of mitigation pathways^{10,11}, it has proven challenging to incorporate SDG performance directly into the design of the pathways themselves³³. Combining climate–economy modelling and multi-objective optimization theory, our analysis presents an approach that enriches the evaluation of mitigation pathways depending on a range of beyond-climate policy objectives.

Relatively balanced mitigation pathways, in terms of implications for the examined SDG indicators, primarily focus on energy supply, AFOLU and industry. It is only for the highest ambition of aligning with a 1.5 °C-compliant pathway that additional mitigation through DAC and/or in transport and buildings is required. While benefiting health objectives more, these two latter sectors perform relatively worse on the other SDG dimensions. SDG-balanced 1.7–2 °C-compatible pathways would not involve many NETs beyond land use. Reliance on such technologies, especially BECCS, is commonly criticised for multiple reasons, primarily owing to uncertainty over availability and uptake and to implications for global land and water use^{15,17}. These criticisms are confirmed by the MSVs in our analysis (Fig. 2), where deep mitigation in energy supply (implying high levels of BECCS)

displays more negative impacts and higher socioeconomic uncertainty. Nevertheless, our SDG-balanced portfolios indicate that, to reach net-zero by 2050, the trade-offs in these technologies weigh similarly to other trade-offs from deep mitigation options in transportation and buildings, while all these options are chosen as a ‘last resort’ in SDG-optimizing portfolios, after exhausting non-NET decarbonization levers in energy supply and industry, as well as large-scale land conservation. We also note that the obtained mitigation potentials are additional to current and national policies, which by itself may drive relevant mitigation levers (such as vehicle electrification; Extended Data Fig. 3). The observation that mitigation in the built environment and transportation performs relatively poorly for most SDG indicators except for health may also hint that real-world mitigation efforts and plans so far may have emphasized health co-benefits, yielding more immediate and locally tangible results^{36,37}.

Despite offering valuable insights into the critical role of IAMs and cross-SDG optimization in guiding climate policy, and although helping decision-makers navigate the complex trade-offs between emissions reductions and broader sustainability implications, our analysis does not come without limitations. For example, the allocation of mitigation efforts relies entirely on the obtained MSVs, which are strongly determined by the model applied²² and the capability to reasonably estimate such values. However, IAMs are often criticized for not including a balanced set of mitigation options in each sector²⁴, implying some sectors may be unjustly (dis-)favoured due to differences in sectoral

granularity. For instance, the IAM applied here does not explicitly reflect heat pumps or building retrofits, thereby potentially overestimating difficulties to building sector abatement, affecting its MSVs. Another limitation is the relatively subjective choice of disaggregated allocation options. Here, we distinguished six representative economic sectors, although within most sectors there may remain large differences in how mitigation options affect SDG-related indicators. The same is true for the choice of SDG objectives and indicators, which is constrained by model capabilities³⁸ and target quantification capacity³⁹. Expanding on these dimensions could enrich the results and conceptual completeness of the study—although it would also amplify the computational requirements and hinder the graphical representation and interpretation of results. Also, separating competing technologies (such as electric versus biofuel-driven mobility) would complicate the implicit assumption that the identified mitigation portfolios are linearly additive. Our analysis has only explored global-level impacts, acknowledging, however, that sectoral burden-sharing decisions are made at smaller jurisdictions, where the balance between sectors and SDG impacts as well as challenges to and feasibility of decarbonization efforts—with strong implications for sectoral prioritization—may vary significantly³¹. For example, optimal sectoral mitigation portfolios may look significantly different in developed relative to developing economies. Finally, the effects of potential climate impacts are not considered, although these may significantly affect the identified MSVs and therefore the Pareto frontier of mitigation portfolios.

It is evident that achieving the ambitious Paris targets and the SDGs requires deep, immediate, cross-sectoral transformations recognizing the complex interdependencies between the two agendas. The SDG umbrella seeks to improve sustainability simultaneously at multiple fronts. Therefore, this analysis can be useful to anticipate trade-offs between mitigation and SDGs and couple mitigation with other policies aiming to address these trade-offs. Despite its limitations, our analysis provides an advanced integrated modelling approach internally considering a wide spectrum of SDG outcomes, allowing policymakers to better design pathways that maximize co-benefits, minimize trade-offs and reinforce resilience against the multifaceted challenges of climate change. It complements emerging literature calling for multidimensional justice perspectives in climate research⁴⁰, showing that more balanced—rather than purely cost-efficient—sectoral burden sharing can yield social co-benefits, such as poverty reduction and health improvement. To overcome some of the identified limitations, future studies could build on our methodology to identify MSVs with alternative models, including structural uncertainty in addition to the parametric uncertainty considered in our analysis via the SSP ranges. Also, studies that further differentiate sectors, capture other SDGs, focus on specific regions and include climate impacts as an additional source affecting MSVs, would be valuable to further strengthen the potential contribution of IAMs to real-world policies by providing policymakers with strategies matching climate mitigation with local priorities, under a range of uncertainties⁴¹.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41558-026-02602-3>.

References

- Report of the Conference of the Parties on Its Twenty-First Session, Held in Paris from 30 November to 13 December 2015. Part One: Proceedings (UNFCCC, 2016).
- Transforming Our World: The 2030 Agenda for Sustainable Development (United Nation, 2015); <https://sdgs.un.org/2030agenda>
- Schmidt Tagomori, I. et al. Climate policy and the SDGs agenda: how does near-term action on nexus SDGs influence the achievement of long-term climate goals?. *Environ. Res. Lett.* **19**, 054001 (2024).
- van Soest, H. L. et al. Analysing interactions among Sustainable Development Goals with integrated assessment models. *Glob. Transit.* **1**, 210–225 (2019).
- von Stechow, C. et al. 2°C and SDGs: united they stand, divided they fall?. *Environ. Res. Lett.* **11**, 034022 (2016).
- Soergel, B. et al. A sustainable development pathway for climate action within the UN 2030 Agenda. *Nat. Clim. Change* **11**, 656–664 (2021).
- Fujimori, S. et al. Measuring the sustainable development implications of climate change mitigation. *Environ. Res. Lett.* **15**, E085004 (2020).
- McCollum, D. L. et al. Energy investment needs for fulfilling the Paris Agreement and achieving the Sustainable Development Goals. *Nat. Energy* **3**, 589–599 (2018).
- Iyer, G. et al. Implications of sustainable development considerations for comparability across nationally determined contributions. *Nat. Clim. Change* **8**, 124–129 (2018).
- Moreno, J. et al. Assessing synergies and trade-offs of diverging Paris-compliant mitigation strategies with long-term SDG objectives. *Glob. Environ. Change* **78**, 102624 (2023).
- Moreno, J. et al. The impacts of decarbonization pathways on Sustainable Development Goals in the European Union. *Commun. Earth Environ.* **5**, 1–14 (2024).
- Hermwille, L. et al. *Ensuring an Effective Global Stocktake with a Sectoral Perspective* (Wuppertal Institute, 2022); <https://doi.org/10.48506/opus-8033>
- van de Ven, D. J. et al. Energy and socioeconomic system transformation through a decade of IPCC-assessed scenarios. *Nat. Clim. Change* **15**, 218–226 (2025).
- Fuhrman, J. et al. Ambitious efforts on residual emissions can reduce CO₂ removal and lower peak temperatures in a net-zero future. *Environ. Res. Lett.* **19**, 064012 (2024).
- Stenzel, F. et al. Irrigation of biomass plantations may globally increase water stress more than climate change. *Nat. Commun.* **12**, 1512 (2021).
- Hirata, A. et al. The choice of land-based climate change mitigation measures influences future global biodiversity loss. *Commun. Earth Environ.* **5**, 259 (2024).
- Hof, C. et al. Bioenergy cropland expansion may offset positive effects of climate change mitigation for global vertebrate diversity. *Proc. Natl Acad. Sci. USA* **115**, 13294–13299 (2018).
- Madhu, K., Pauliuk, S., Dhathri, S. & Creutzig, F. Understanding environmental trade-offs and resource demand of direct air capture technologies through comparative life-cycle assessment. *Nat. Energy* **6**, 1035–1044 (2021).
- Rogelj, J. et al. Scenarios towards limiting global mean temperature increase below 1.5 °C. *Nat. Clim. Change* **8**, 325–332 (2018).
- Edelenbosch, O. Y. et al. Reducing sectoral hard-to-abate emissions to limit reliance on carbon dioxide removal. *Nat. Clim. Change* **14**, 715–722 (2024).
- Nikas, A., Doukas, H. & Papandreou, A. in *Understanding Risks and Uncertainties in Energy and Climate Policy: Multidisciplinary Methods and Tools for a Low Carbon Society* (eds Doukas, H., Flamos, A. & Lieu, J.) (Springer, 2019); <https://doi.org/10.1007/978-3-030-03152-7>
- Dekker, M. M. et al. Identifying energy model fingerprints in mitigation scenarios. *Nat. Energy* **8**, 1395–1404 (2023).
- Kowarsch, M. in *Understanding and Evaluating the IAM-Based Economics* (Springer, 2016); https://doi.org/10.1007/978-3-319-43281-6_7

24. Gambhir, A., Ganguly, G. & Mittal, S. Climate change mitigation scenario databases should incorporate more non-IAM pathways. *Joule* **6**, 2663–2667 (2022).
25. Pindyck, R. S. The use and misuse of models for climate policy. *Rev. Environ. Econ. Policy* **11**, 100–114 (2017).
26. Sampedro, J. et al. Health co-benefits and mitigation costs as per the Paris Agreement under different technological pathways for energy supply. *Environ. Int.* **136**, 105513 (2020).
27. Soergel, B. et al. Combining ambitious climate policies with efforts to eradicate poverty. *Nat. Commun.* **12**, 2342 (2021).
28. Ambrósio, G., Doelman, J. C., Schipper, A. M., Stehfest, E. & van Vuuren, D. Global sustainability scenarios lead to regionally different outcomes for terrestrial biodiversity. *Environ. Res. Lett.* **19**, 104055 (2024).
29. Séférian, R., Rocher, M., Guivarch, C. & Colin, J. Constraints on biomass energy deployment in mitigation pathways: the case of water scarcity. *Environ. Res. Lett.* **13**, 054011 (2018).
30. *Emissions Gap Report 2023: Broken Record—Temperatures Hit New Highs, Yet World Fails to Cut Emissions (Again)* (United Nations Environment Programme, 2023); <https://doi.org/10.59117/20.500.11822/43922>
31. van de Ven, D.-J. et al. A multimodel analysis of post-Glasgow climate targets and feasibility challenges. *Nat. Clim. Change* **13**, 570–578 (2023).
32. Forouli, A., Nikas, A., Van de Ven, D. J., Sampedro, J. & Doukas, H. A multiple-uncertainty analysis framework for integrated assessment modelling of several sustainable development goals. *Environ. Model. Softw.* **131**, 104795 (2020).
33. Nikas, A. Projecting progress in sustainable development goals vis-à-vis climate action in climate–economy models. *PLoS Clim.* **3**, e0000449 (2024).
34. Riahi, K. et al. The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: an overview. *Glob. Environ. Change* **42**, 153–168 (2017).
35. Gupta, A. in *Handbook of Environmental and Sustainable Finance* (Elsevier, 2016); <https://doi.org/10.1016/B978-0-12-803615-0.00001-7>
36. Shoemaker, J. K., Schrag, D. P., Molina, M. J. & Ramanathan, V. What role for short-lived climate pollutants in mitigation policy?. *Science* **342**, 1323–1324 (2013).
37. Fekete, H. et al. A review of successful climate change mitigation policies in major emitting economies and the potential of global replication. *Renew. Sustain. Energy Rev.* **137**, 110602 (2021).
38. Koasidis, K., Koutsellis, T., Xexakis, G., Nikas, A. & Doukas, H. Understanding expectations from and capabilities of climate-economy models for measuring the impact of crises on sustainability. *J. Clean. Prod.* **414**, 137585 (2023).
39. van Vuuren, D. P. et al. Defining a sustainable development target space for 2030 and 2050. *One Earth* **5**, 142–156 (2022).
40. Zimm, C. et al. Justice considerations in climate research. *Nat. Clim. Change* **14**, 22–30 (2024).
41. Peng, W. et al. Climate policy models need to get real about people—here’s how. *Nature* **594**, 174–176 (2021).
42. Byers, E. et al. AR6 scenarios database. Zenodo <https://doi.org/10.5281/ZENODO.5886912> (2022).
43. Fuss, S. et al. Negative emissions—part 2: costs, potentials and side effects. *Environ. Res. Lett.* **13**, 063002 (2018).

Publisher’s note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

Methods

The methodological framework of this study is broken down into seven steps (graphical overview in Fig. 1): (1) defining sectoral mitigation potential towards 2050, (2) creating incremental scenarios with varying sectoral effort, (3) describing the IAM environment applied to run scenarios, (4) identifying and fully describing the applied assumptions for five SDG indicators, (5) synthesis of using multi-objective integer programming (MOIP) to find optimal portfolios, (6) selecting and validating a SDG-balanced sectoral portfolios and (7) running and comparing least-cost mitigation scenarios.

Sectoral mitigation potential (step 1)

Estimates of sectoral contributions to GHG mitigation vary widely across studies due to differences in scenario design, models and assumptions about future demand and technological change. Recent assessments (up to 2024) consistently find that currently implemented and planned policies ('current policies') are insufficient to meet the Paris Agreement, leading instead to around 2–3 °C of warming by 2100^{30,31}. However, compared with scenarios aimed at deep mitigation (for example, 1.5 °C or 2 °C pathways, often relying on uniform global carbon pricing), current policies scenarios offer more reliable insights into the short- to medium-term sectoral distribution of emissions⁴⁴. This is because many policies explicitly or implicitly target specific sectors and the inclusion of national commitments (such as NDCs) better reflects regional differences in sectoral emissions. By contrast, scenarios based on global carbon pricing are better suited to illustrating the overall mitigation potential by sector.

To estimate the maximum sectoral mitigation potentials, this study draws on the AR6 scenario database⁴² to construct sectoral emission ranges for a NPi baseline and for scenarios compatible with 2 °C, 1.7 °C and 1.5 °C warming up to 2050. These ranges are combined with observed sectoral emissions data up to 2020 (EDGAR⁴⁴) to derive a sectoral decomposition of the NPi baseline and corresponding mitigation ranges for each sector. Climate-target-consistent scenarios are selected based on median projected warming by 2100, with additional inclusion of non-full-century scenarios using 2050 CO₂ emission ranges consistent with the selected temperature classes. NPi scenarios are identified using policy-related scenario tags (NPi, CurP, NDCi, NDCp, Current policies, CP_intensity, CP_Price, NDC_Intensity and NDC_Price) and filtered by plausible 2050 CO₂ emissions to exclude pathways already aligned with 2 °C targets or outside realistic current-policy outcomes⁴⁵. Sectoral emissions are extracted for 2010, 2030 and 2050, with harmonized sector definitions and model-averaged results to avoid model dominance. Where scenario coverage is limited (for example, DAC⁴³), literature-based estimates are applied. Sectoral pathways are anchored to observed 2020 emissions and interpolated forwards. The resulting mitigation potentials differ substantially across sectors, reflecting differences in current emissions levels and the feasibility of reaching net-negative emissions, which are possible primarily in the energy supply and AFOLU sectors. Minimum sectoral emission benchmarks are defined using the lower (10%) bound of 1.5 °C-compatible pathways in 2050, with 2030 values interpolated from observed 2020 emissions to ensure consistency with recent trends. The results of this exercise can be found in Table 1 and Extended Data Fig. 1.

Incremental scenarios and emission budgets (step 2)

As our analysis assesses sectoral mitigation in cumulative-emission terms, emission pathways from the previous step are converted into cumulative sectoral potentials. We calculate the area between each sector's NPi mean pathway $E(\text{NPi})$ and the minimal benchmark $E(\text{min})$. As shown in Extended Data Fig. 4, top, the area between $E(\text{NPi})$ and $E(\text{min})$ is divided into four components (A – D), with formulas differing slightly depending on whether $E(\text{NPi})$ rises or falls after 2030. For Extended Data Fig. 4, top,

$$A = (E(\text{NPi})_{2030} - E(\text{min})_{2030}) \times (2030 - 2020) \times 0.5$$

$$B = (\text{MAX}(E(\text{NPi})_{2030}, E(\text{NPi})_{2050}) - \text{MIN}(E(\text{NPi})_{2030}, E(\text{NPi})_{2050})) \times (2050 - 2030) \times 0.5$$

$$C = (\text{MIN}(E(\text{NPi})_{2030}, E(\text{NPi})_{2050}) - E(\text{min})_{2030}) \times (2050 - 2030)$$

$$D = (E(\text{min})_{2030} - E(\text{min})_{2050}) \times (2050 - 2030) \times 0.5.$$

Where A represents cumulative reductions from 2020 to 2030 and $B + C + D$ represent reductions from 2030 to 2050. The total sectoral savings ($A + B + C + D$) are split into 5-Gt blocks to create multiple incremental emission pathways per sector (Extended Data Fig. 4, bottom).

This yields sector-specific mitigation 'budgets' and varying numbers of incremental scenarios (Extended Data Table 2). Each 5-Gt block is run through GCAM, and five SDG indicators (defined below) are evaluated across all 135 scenarios to estimate marginal impacts of mitigation scaling (MSVs; Fig. 1). The procedure is repeated for all five SSPs to reflect parametric uncertainty.

To isolate the sectoral effects, only emissions in the target sector vary in each run and all others are fixed at NPi levels via sectoral carbon prices. This prevents cross-sector spillovers from altering emissions elsewhere (for example, to energy supply through electrification or to AFOLU through biofuels), ensuring that any changes in other sectors (for example, to compensate for increased electricity or biofuel demand) are attributed to the incremental mitigation in the focal sector.

Finally, we define emission budgets for pathways compatible with 2 °C, 1.7 °C and 1.5 °C warming, to which the sectoral contributions from Table 1 would have to add up. For consistency with sectoral potentials, we use the mean sectoral estimates across all sectors to these budgets (Table 1). The required cumulative reductions relative to the NPi baseline are ~300 Gt (2 °C), ~415 Gt (1.7 °C) and ~560 Gt (1.5 °C).

IAM environment (step 3)

We apply the GCAM⁴⁶ to run multiple incremental scenarios for each of the five SSPs, increasing mitigation in one sector in steps of 5 Gt up to their maximum mitigation potential (Table 1). At each level, we identify the relative impacts in five SDG indicators.

GCAM is a dynamic-recursive, technology-rich model representing the economy, energy, land use, water and climate. It is used to assess mitigation policies such as carbon taxes, trading, regulations and accelerated technology deployment. Population and labour-productivity assumptions drive energy and land-use systems, with detailed technology options for energy services, agriculture, forestry and land cover. Running up to 2100 in 5-year steps, GCAM has been widely applied to evaluate emerging energy technologies and GHG impacts of policies, including CO₂ capture and storage, bioenergy, hydrogen, nuclear, renewables and end-use technologies in buildings, industry and transport.

We use GCAM version 7.2, which incorporates updated SSP projections for population, GDP and income elasticities (v3.0)⁴⁷ and includes income-decile detail for residential energy⁴⁸ and food demand⁴⁹. To separate sectoral emissions, we introduced adjustments in the emission accountability of bioenergy use, which does not affect model dynamics, but only the accountability of positive and negative emissions. This enables consistent allocation of CO₂ emissions and removals across six sectors: industry, buildings, transport, AFOLU, energy supply and DAC. Consistent with other GCAM applications, land-use carbon flows are priced or subsidized at 20% of the standard carbon price for energy and industrial CO₂, reflecting higher uncertainty in sequestration from land conservation and afforestation and the correspondingly lower effective payments to land owners.

SDG indicator choice and definition (step 4)

From GCAM outputs, we derive indicators for five SDGs selected because they (1) represent different sustainability dimensions, (2) are driven by IAM outputs rather than inputs and (3) are mutually non-correlated. Some indicators require post-processing within the GCAM ecosystem; this is automated via the GCAM_SDG generator tool⁵⁰, which converts GCAM outputs into the SDG indicators defined below. The indicators are

- Income expenditure share for basic needs (home energy and food) for SDG 1: ‘no poverty’;
- Premature mortality due to ambient air pollution for SDG 3: ‘good health and well-being’;
- Physical water scarcity index for SDG 6: ‘clean water and sanitation’;
- GDP per capita growth for SDG 8: ‘decent work and economic growth’;
- Net PSL as a biodiversity indicator for SDG 15: ‘life on land’.

We assess impacts on these indicators at each incremental mitigation level, deriving marginal SDG values (MSVs)⁷ separately for each SSP, always relative to the SSP-specific baseline. Sectoral mitigation targets (NPI and incremental 5-Gt steps) are identical across SSPs, but carbon prices and non-emission outcomes differ due to SSP parameterization (see Extended Data Fig. 2 for NPI). MSVs represent average impacts over 2025–2050 for all indicators except PSL (SDG 15), which is measured as the 2050 value relative to 2020.

Mitigation efforts designed to meet current climate pledges have far-reaching implications across the full spectrum of the United Nations 2030 Agenda that extend well beyond its formal deadline in 2030, emphasizing that only the 2030 outcomes might not be sufficient to capture the strategic shift towards mid- to long-term planning¹⁰. It is reasonable to expect that subsequent global development frameworks will retain these goals, as occurred in the transition from the Millennium Development Goals to the SDGs. In addition, many mitigation measures involve long-lived capital investments and structural transformations whose effects unfold over several decades, implying that outcomes observed by 2030 reflect only the early stages of these transitions⁵¹. This analysis therefore focuses on projections until 2050 to assess the long-term consequences of present mitigation decisions. The specific method for defining each indicator is listed below.

SDG 1: no poverty. To assess mitigation impacts on poverty, we measure household expenditures on residential energy and food relative to average income, capturing the pressure to meet basic needs. The indicator is calculated for all representative households in GCAM, defined as 10 income deciles across 32 socioeconomic regions (320 households in total). The results are averaged across households, weighted by their population shares. Although this includes non-poor households, absolute-poverty thresholds are subjective so we use a relative-poverty measure. As poorer households have lower incomes (denominators), they show the highest expenditure shares and largely drive indicator variation. Thus, small global changes may still reflect substantial impacts among the poorest groups. The income expenditure share for basic needs (IESBN) is calculated as

$$\text{IESBN}_{t,s} = \left(\frac{\sum_r \sum_h \left(\frac{E_{t,s}}{I_{t,s}} + \frac{F_{t,s}}{I_{t,s}} \right) \times \text{pop}_{t,s}}{\sum_r \sum_h \text{pop}_{t,s}} \right),$$

where E reflects home energy expenditure (in US\$), F is food expenditure (in US\$), I is income (in US\$), r is the socioeconomic regions in GCAM, h is the representative households (income deciles) within those regions, pop is the population of the representative household, t is the model year and s is the scenario.

SDG 3: good health and well-being. SDG 3 is represented as avoided premature mortality from long-term exposure to fine particulate matter (PM_{2.5}) and tropospheric ozone (O₃). The estimates were produced with *rfasst*, a tool that quantifies health and agricultural impacts of air pollution under alternative scenarios⁵² and emulates the well-established TM5-FASST model⁵³.

Premature mortality is calculated using a population-attributable fraction approach. Mortality (mort) for cause c , time t , region r and pollutant j equals the product of the baseline mortality rate, the change in relative risk (RR) from changes in population-weighted pollutant concentration and the exposed population

$$\text{Mort}_{c,t,r,j} = \text{mort}_{t,c,r,j} \times \left(\frac{\text{RR}_{t,c,j} - 1}{\text{RR}_{t,c,j}} \right) \times \text{pop}_{t,r}.$$

Mortalities are estimated for six causes: stroke, ischaemic heart disease, chronic obstructive pulmonary disease, acute lower respiratory infections, lung cancer and diabetes mellitus type II. Ischaemic heart disease and stroke use age-specific mortality rates and relative risks; chronic obstructive pulmonary disease, acute lower respiratory infections, lung cancer and diabetes mellitus type II are estimated for the population aged >25 years.

Cause-specific baseline mortality rates are derived from Global Burden of Disease data and population statistics, with future changes based on World Health Organization projections. For PM_{2.5}, relative risks use integrated exposure–response functions from Global Burden of Disease 2017⁵⁴, which include age-specific parameters and diabetes type II. For O₃, relative risks follow the exposure–response functions of Jerrett et al.⁵⁵.

SDG 6: clean water a sanitation. The water module of GCAM is structured in 235 basins⁴⁶. Water withdrawals—defined as water diverted from surface or groundwater sources—are estimated for six sectors: agriculture, electricity generation, industrial manufacturing, primary energy production, livestock and municipal use.

Water supply distinguishes three freshwater sources: renewable water, non-renewable groundwater and desalinated water. Renewable water is naturally replenished through surface runoff and subsurface processes and is determined by streamflow, baseflow, reservoir storage and environmental flow requirements in each basin⁵⁶.

Physical water scarcity is quantified as the ratio of water withdrawals to renewable water supply in each basin⁵⁷, also referred to as the water stress index in previous studies⁵⁸. A single scenario value is calculated as the basin index average weighted by renewable water withdrawals in the baseline year (2015). The index is defined as

$$I_{s,b} = \frac{W_{s,b}}{S_{s,b}} \quad \text{and} \quad I_s = \frac{\sum_{b=1}^b (I_{s,b} \times W_{s,b,2015})}{\sum_{b=1}^b W_{s,b,2015}},$$

where I is the water scarcity index (dimensionless), W is water withdrawal (km³), S is the renewable water supply (km³), b is basins and s is scenarios.

SDG 8: decent work and economic growth. The SDG 8 indicator is the global average GDP per capita growth, obtained directly from GCAM’s macroeconomic module⁵⁹. SSP-defined GDP paths apply to the NPI reference scenario, whereas the macroeconomic module is activated for mitigation scenarios. The indicator measures GDP impacts relative to the SSP-specific NPI baseline.

SDG 15: life on land. The GCAM land module represents 384 land–water regions (LUTs) and reports land allocation across 43 uses per LUT. It models supply, demand and land use for food, feed, fibre, forestry and bioenergy⁴⁶. Land is classified as managed or unmanaged; unmanaged

categories include unmanaged forests, pasture, shrubland, grassland, tundra and other land (for example, rock, ice and desert).

The SDG 15 indicator is net PSL, representing plant and vertebrate species committed to extinction due to habitat loss from LUC from unmanaged to managed land. PSL accounts for species threat levels, endemism and adaptive capacity^{60,61}.

First, GCAM land-use projections are downscaled with the Demeter model⁶² to $0.5^\circ \times 0.5^\circ$ resolution and aggregated to ecoregions, the spatial unit required for biodiversity assessment⁶¹. For each ecoregion and land use, LUC equals the area difference between 2020 and 2050.

Second, PSL is calculated using ecoregion-based LUC and aggregated characterization factors (CFs), which express species loss per hectare for five managed land uses (forestry, cropland (irrigated and rain fed), pasture land and urban land) at different management intensities across 804 terrestrial ecoregions. Extended Data Table 3 presents the mapping between GCAM land uses and CF categories. CFs reflect the combined losses across mammals, birds, amphibians, reptiles and plants and are derived from ref. 61.

The PSL indicator of each scenario is estimated with the following equations:

$$LUC_{l,r} = A_{l,r,2050} - A_{l,r,2020}$$

$$PSL_{l,r} = LUC_{l,r} \times CF_{l,r}$$

$$PSL = \sum_l \sum_r PSL_{l,r}$$

where A is the area in hectares, LUC is the LUC between 2020 and 2050 (in hectares), CF is the aggregated CF (species loss per hectares), PSL is the total net potential species loss, l is managed land uses and r is ecoregions.

Robust portfolio analysis (step 5)

A key difference from a standalone IAM exercise is that the MOIP set up explicitly identifies the performance that can be optimized simultaneously across all dimensions. Rather than a single deterministic optimum, it produces a set of Pareto-optimal (non-dominated) solutions showing the achievable trade-offs without worsening other objectives.

The MOIP algorithm processes the marginal SDG values (MSVs) derived above to identify combinations of sectoral mitigation levels that yield the Pareto-optimal performance across the five sustainability dimensions, subject to the mitigation required for each temperature target (three separate runs). We solve the MOIP using AUMECON-Py⁶³, an open-source implementation and expansion of the AUGMECON-R algorithm⁶⁴ building on the ϵ -constraint family and the augmented (AUGMECON) versions of optimization methods.

The formulation of each one of the three MOIP problems is as follows:

Max {economy} and

min {poverty, mortalities, water scarcity, PSL},

subject to $\sum_{i=1}^5$ sectoral mitigation effort _{i}

$\geq$ total mitigation effort required – 5 GtCO₂,

$\sum_{i=1}^5$ sectoral mitigation effort _{i} $\leq$ total mitigation effort required + 5 GtCO₂.

A pay-off table is constructed containing the performance of each sectoral mitigation level (MSVs from GCAM; Fig. 2) across sustainability

dimensions. As the results differ by SSP, we use the mean across the five SSP runs, with SSP variability addressed in the uncertainty analysis. For consistency, the minimization objectives and the ‘greater-than’ constraint are converted to maximization form by sign inversion, and a ± 5 GtCO₂ tolerance is applied to ensure that the solutions calculated are in line with the targets selected and avoid under- or overperformance.

To account for the impact of SSP-induced uncertainty in the outcomes of the optimization process, we also calculate the standard deviation from the mean values—which were used to form the pay-off table—for each pair of sector and sustainability indicators. In this process we exclude outliers by calculating Tukey’s fences⁶⁵ and omitting any values not in the range ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$), where $Q1$ and $Q3$ are the first and third quartiles and IQR is the interquartile range ($Q3 - Q1$). Assuming normal distributions, we then perform 10,000 Monte Carlo iterations to generate a broader set of Pareto-optimal allocations reflecting stochastic SSP variability. Robustness is assessed by the frequency with which each sectoral allocation mix appears across iterations.

Selection and model assessment of an ‘SDG-balanced’ portfolio (step 6)

In the final step, we select one portfolio per climate target that performs well across all five SDGs, then re-run GCAM with this sectoral mitigation mix to validate the approach and illustrate the characteristics of an ‘SDG-balanced’ scenario.

The selection of this portfolio (P^*) is driven by the relative performance (O) on every SDG compared with the performance of all other portfolios in the same optimization problem (that is, temperature target), applying the following formula:

$$P^* = \min \sum_{SDG} \left(\frac{\max_{SDG} O_p - O_p}{\max_{SDG} O_p - \min_{SDG} O_p} \right).$$

Robustness—defined as the number of Monte Carlo iterations in which a portfolio appears on the Pareto front—is used only as a validation check. We consider portfolios robust if they appear in ≥ 100 iterations ($\geq 1\%$ of runs). Robustness is not included in the selection formula because it correlates with SDGs that show low SSP variability (for example, PSL; Fig. 2), which would bias selection.

The P^* portfolios for each temperature target are then translated back into sectoral emissions targets, applying the same temporal structure of mitigation as used for the incremental scenarios per sector (Extended Data Figs. 1 and 4), with the relative level of mitigation in each sector depending on its share in the P^* . These sectoral emission targets are implemented into GCAM, in this instance for all six sectors at once in the same scenario (in contrast to the incremental scenarios). The resulting scenario outcomes are then again evaluated with the GCAM_SDG generator tool to validate the SDG outcomes of the selected portfolio (‘sdg balanced (model)’ results in Fig. 4c).

Selected P^* portfolios are converted into sectoral emission targets using the same temporal mitigation structure as the incremental sector scenarios (Extended Data Figs. 1 and 4), scaled by each sector’s share in P^* . These targets are implemented simultaneously across all six sectors in GCAM (unlike the incremental runs), allowing to validate the relevance of intersectoral dependencies. Scenario outcomes are then evaluated with the GCAM_SDG generator tool to validate SDG results (‘sdg balanced (model)’ in Fig. 4c).

Defining and running comparable mitigation scenarios based on least-cost allocation (step 7)

To compare with SDG-balanced scenarios, we also ran least-cost scenarios for each temperature target by aggregating sectoral CO₂ emissions

at each model period and applying a single total mitigation constraint without predefined sectoral allocations. In this standard IAM set up, mitigation is allocated by sector under a uniform CO₂ price that rises until the target is met.

To ensure comparability, we align the temporal mitigation profile as closely as possible. In SDG-balanced scenarios, the price applied to LUC carbon flows (typically 20% of the standard carbon price, as noted in ‘IAM environment’ section) follows a distinct pattern—very low initially and much higher by 2050—reflecting AFOLU mitigation shares and sectoral abatement costs. Compared with the weighted-average carbon price for fossil CO₂, this corresponds to relative prices of ~5% in 2030, 20% in 2040 and 30% in 2050, which we replicate in the least-cost runs. Outcomes from the least-cost scenarios are evaluated with the GCAM_SDG generator tool and compared with the SDG-balanced results (Fig. 4c).

Data availability

The datasets generated during and analysed in the current study, as well as the full details of the NPi, least-cost and post-Pareto SDG-balanced scenarios (in IPCC-style format) are available via Zenodo at <https://doi.org/10.5281/zenodo.18633223> (ref. 66). Model input data (equations, assumptions and parameters) are included in the online model repository referred to in the Code availability statement. The latest projections of the net income distribution used for this analysis are publicly available from Zenodo at <https://doi.org/10.5281/zenodo.7474549> (refs. 67).

Code availability

The analysis has been developed using an enhanced version of the open-source GCAM and is available via GitHub at https://github.com/bc3LC/gcam-core/tree/bioaccounting_7p1. A detailed documentation for all these input assumptions used in the GCAM model is available via GitHub at <https://github.com/JGCRI/gcam-doc>. The post-processing code to calculate the mentioned SDG outcomes from GCAM scenarios is available via GitHub at https://github.com/bc3LC/gcam_sdg.

References

44. Crippa, M. et al. GHG emissions of all world countries (Publications Office of the European Union, 2023); <https://doi.org/10.2760/953322>
45. Hausfather, Z. & Peters, G. P. Emissions—the ‘business as usual’ story is misleading. *Nature* **577**, 618–620 (2020).
46. Calvin, K. et al. GCAM v5.1: representing the linkages between energy, water, land, climate, and economic systems. *Geosci. Model Dev.* **12**, 677–698 (2019).
47. Zhao, X. et al. *Core Model Proposal #399: Updating the SSP Database (v3.0) (Population, GDP, and Labor Force) and Labor Productivity (KLEM)* (US Department of Energy Office of Scientific and Technical Information, 2024); <https://doi.org/10.2172/2999947>
48. Sampedro, J. et al. Residential energy demand, emissions, and expenditures at regional and income-decile level for alternative futures. *Environ. Res. Lett.* **19**, 084031 (2024).
49. Waldhoff, S. et al. Analyzing the distributional impacts of global change on food access and availability in a multi-sector dynamics, human–earth system model. In *AGU Fall Meeting Abstracts GC22B-06* (AGU, 2023).
50. *gcam_sdg*. GitHub https://github.com/bc3LC/gcam_sdg (2025).
51. Bertram, C. et al. Carbon lock-in through capital stock inertia associated with weak near-term climate policies. *Technol. Forecast. Soc. Change* **90**, 62–72 (2015).
52. Sampedro, J. et al. *rfasst*: an R tool to estimate air pollution impacts on health and agriculture. *J. Open Source Softw.* **7**, 3820 (2022).
53. Van Dingenen, R. et al. TM5-FASST: a global atmospheric source-receptor model for rapid impact analysis of emission changes on air quality and short-lived climate pollutants. *Atmos. Chem. Phys.* **18**, 1–55 (2018).
54. Stanaway, J. D. et al. Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet* **392**, 1923–1994 (2018).
55. Jerrett, M. et al. Long-term ozone exposure and mortality. *N. Engl. J. Med.* **360**, 1085–1095 (2009).
56. Kim, S. H. et al. Balancing global water availability and use at basin scale in an integrated assessment model. *Clim. Change* **136**, 217–231 (2016).
57. Birnbaum, A., Lamontagne, J., Wild, T., Dolan, F. & Yarlagadda, B. Drivers of future physical water scarcity and its economic impacts in Latin America and the Caribbean. *Earth's Future* **10**, e2022EF002764 (2022).
58. Byers, E. et al. Global exposure and vulnerability to multi-sector development and climate change hotspots. *Environ. Res. Lett.* **13**, 055012 (2018).
59. Edmonds, J. et al. Sensitivity of future regional and global energy markets and macroeconomic activity to a hypothetical global energy market disruption. *iScience* **28**, 111449 (2025).
60. Chaudhary, A., Veronesi, F., de Baan, L. & Hellweg, S. Quantifying land use impacts on biodiversity: combining species–area models and vulnerability indicators. *Environ. Sci. Technol.* **49**, 9987–9995 (2015).
61. Chaudhary, A. & Brooks, T. M. Land use intensity-specific global characterization factors to assess product biodiversity footprints. *Environ. Sci. Technol.* **52**, 5094–5104 (2018).
62. Vernon, C. R. et al. Demeter—a land use and land cover change disaggregation model. *J. Open Res. Softw.* **6**, 15 (2018).
63. Forouli, A. et al. AUGMECON-Py: a Python framework for multi-objective linear optimisation under uncertainty. *SoftwareX* **20**, 101220 (2022).
64. Nikas, A., Fountoulakis, A., Forouli, A. & Doukas, H. A robust augmented ϵ -constraint method (AUGMECON-R) for finding exact solutions of multi-objective linear programming problems. *Oper. Res.* **22**, 1291–1332 (2022).
65. Hoaglin, D. C., Iglewicz, B. & Tukey, J. W. Performance of some resistant rules for outlier labeling. *J. Am. Stat. Assoc.* **81**, 991–999 (1986).
66. Van de Ven, D.-J., Rod s-Bachs, C., Rouhette, T. & Koasidis, K. Intermediary and final data output from: ‘From least-cost to SDG-optimal sectoral allocation of Paris-compatible mitigation effort’. Zenodo <https://doi.org/10.5281/zenodo.18633223> (2026).
67. Narayan, K. B., O’Neill, B. C., Waldhoff, S. & Tebaldi, C. Data supplement to “Non-parametric projections of national income distribution consistent with the SSPs”. Zenodo <https://doi.org/10.5281/zenodo.7474549> (2022).

Acknowledgements

We thank the Horizon Europe European Commission Projects ‘IAM COMPACT’ (grant no. 101056306 to D.-J.V.d.V., C.R.-B., T.H., R.H., J.S., A.N., N.F. and K.K.), DIAMOND (grant no. 101081179 to D.-J.V.d.V., C.R.-B., T.H., R.H., J.S., A.N., N.F. and K.K.) and ACCLIMATE (grant no. 101184374 to D.-J.V.d.V., T.H., N.F. and A.N.) and the Horizon 2020 European Commission Project ‘NDC ASPECTS’ (grant no. 101003866 to D.-J.V.d.V. and T.H.). G.I. and X.Z. are also affiliated with Pacific Northwest National Laboratory, which did not provide specific support for this paper. The views and opinions expressed in this paper are those of the authors alone and do not necessarily state or reflect those of the affiliated organizations or the US Government, and no official endorsement should be inferred.

Author contributions

D.-J.V.d.V., K.K. and A.N. coordinated the study design. D.-J.V.d.V., C.R.-B., T.R., R.H., J.S., A.N., N.F. and K.K. were responsible for the compilation of the analysis and figures. D.-J.V.d.V. coordinated the conception and writing of the paper with notable contributions and feedback from all other authors, including X.Z., A.C., G.I. and J.M.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41558-026-02602-3>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41558-026-02602-3>.

Correspondence and requests for materials should be addressed to Dirk-Jan Van de Ven.

Peer review information *Nature Climate Change* thanks Shu Zhang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Extended Data Table 1 | Sectoral decomposition and SDG performance of selected SDG-balanced mitigation pathways

scenarios		2C SDG-Balanced	1.7C SDG-balanced	1.5C SDG-balanced	Unit
Cumulative (2020-2050) mitigation by sector relative to NPi	Industry	70	160	180	Gt CO ₂
	Buildings	0	0	5	Gt CO ₂
	Transport.	0	0	20	Gt CO ₂
	AFOLU	55	70	85	Gt CO ₂
	Energy Supply	180	190	245	Gt CO ₂
	DAC	0	0	20	Gt CO ₂
Performance by objective* by adding up MSVs * Mitigation, within solution space (brackets)	Poverty	0.47 (0.24–1.00)	0.67 (0.37–1.12)	0.92 (0.64–1.32)	% of income
	Health	82.8 (54–101)	115 (76–130)	138 (119–151)	avoided mort.*1k
	Water	-2.7 (-3.3 – +2.5)	-2.7 (-3.1 – +0.9)	-1.7 (-2 – +0.8)	Δ index
	Economy	0.37 (0.28–1.02)	0.54 (0.43–1.19)	0.87 (0.69–1.36)	% pcGDP reduct.
	Land	-165 (-273 – +17)	-216 (-272 – +13)	-264 (-272 – 0)	Δ PSL
Performance by objective* from model simulation (% diff with above value)	Poverty	0.48 (+2.5%)	0.70 (+3.9%)	1.01 (+10.5%)	% of income
	Health	85.4 (+3.1%)	119.6 (+4.3%)	149.7 (+8.7%)	avoided mort.*1k
	Water	-2.6 (-6.2%)	-2.5 (-9.4%)	-1.7 (-1.1%)	Δ index
	Economy	0.37 (-0.6%)	0.54 (-0.7%)	0.89 (+2.5%)	% pcGDP reduct.
	Land	-174 (+5.3%)	-223 (+3.0%)	-282 (+6.4%)	Δ PSL

“Performance” is measured by the average outcome of the SDG metric (see Methods section) over the 5 SSP runs Sectoral decomposition of selected SDG-balanced mitigation pathways by temperature target, and performance over 5 SDG indicators calculated both by multiplying mitigation with MSVs, and by re-calculating the impact from model-driven simulation of SDG-balanced portfolios, for validation purposes.

Extended Data Table 2 | Cumulative sectoral emission saving potentials

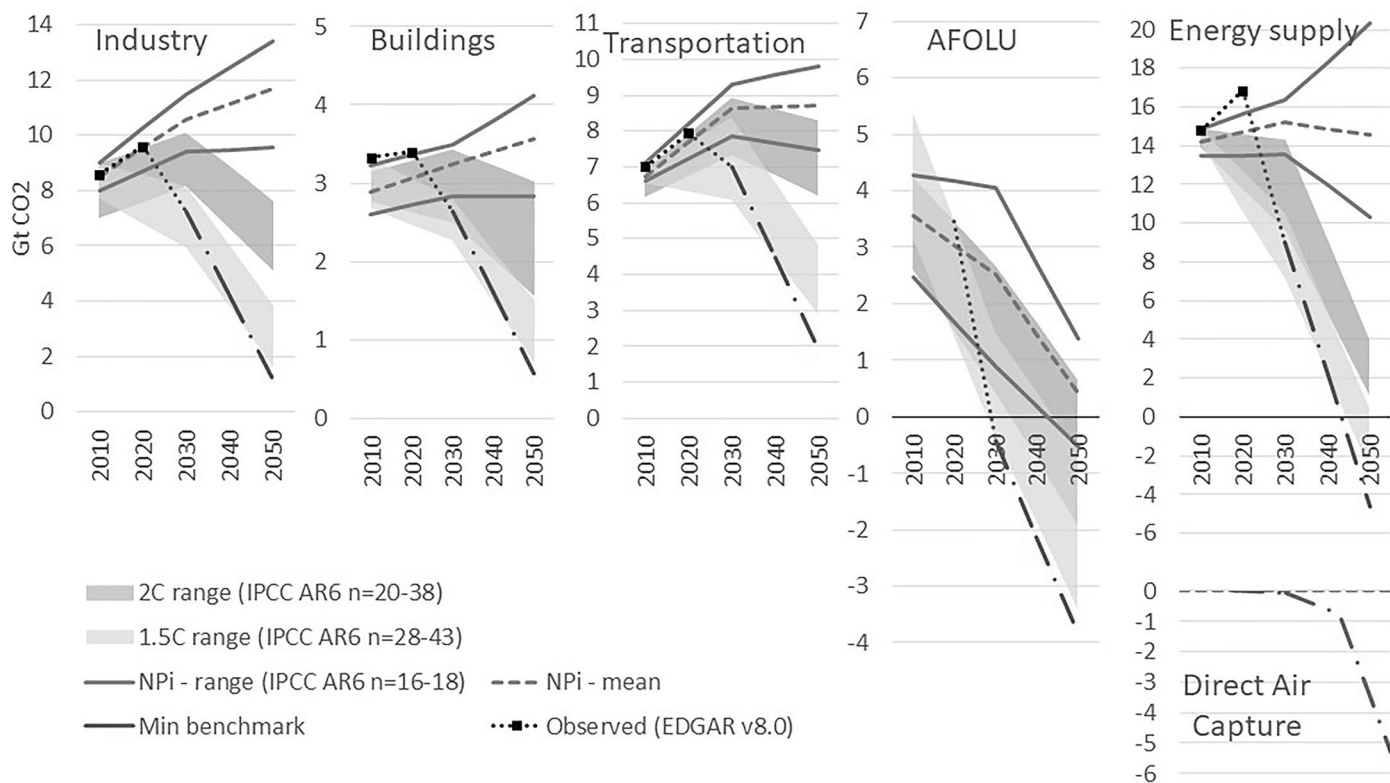
Sector	Area A (2020-2030)	Area B+C+D (2030-2050)	Total (2020-2050)	# 5 Gt blocks (rounded)
Industry	22.5	158.2	180.8	36
Buildings	3.1	36.3	39.3	8
Transportation	3.4	60.7	64.2	13
AFOLU	14.7	71.9	86.7	17
Energy Supply	27.1	242.0	269.1	54
DAC	0.3	34.3	34.6	7
<i>Total</i>	<i>71.1</i>	<i>601.6</i>	<i>672.7</i>	<i>135</i>

Cumulative sectoral emission saving potentials (NPi vs. minimum benchmark) and applied incremental scenarios by sector.

Extended Data Table 3 | Land use type mapping for PSL indicator

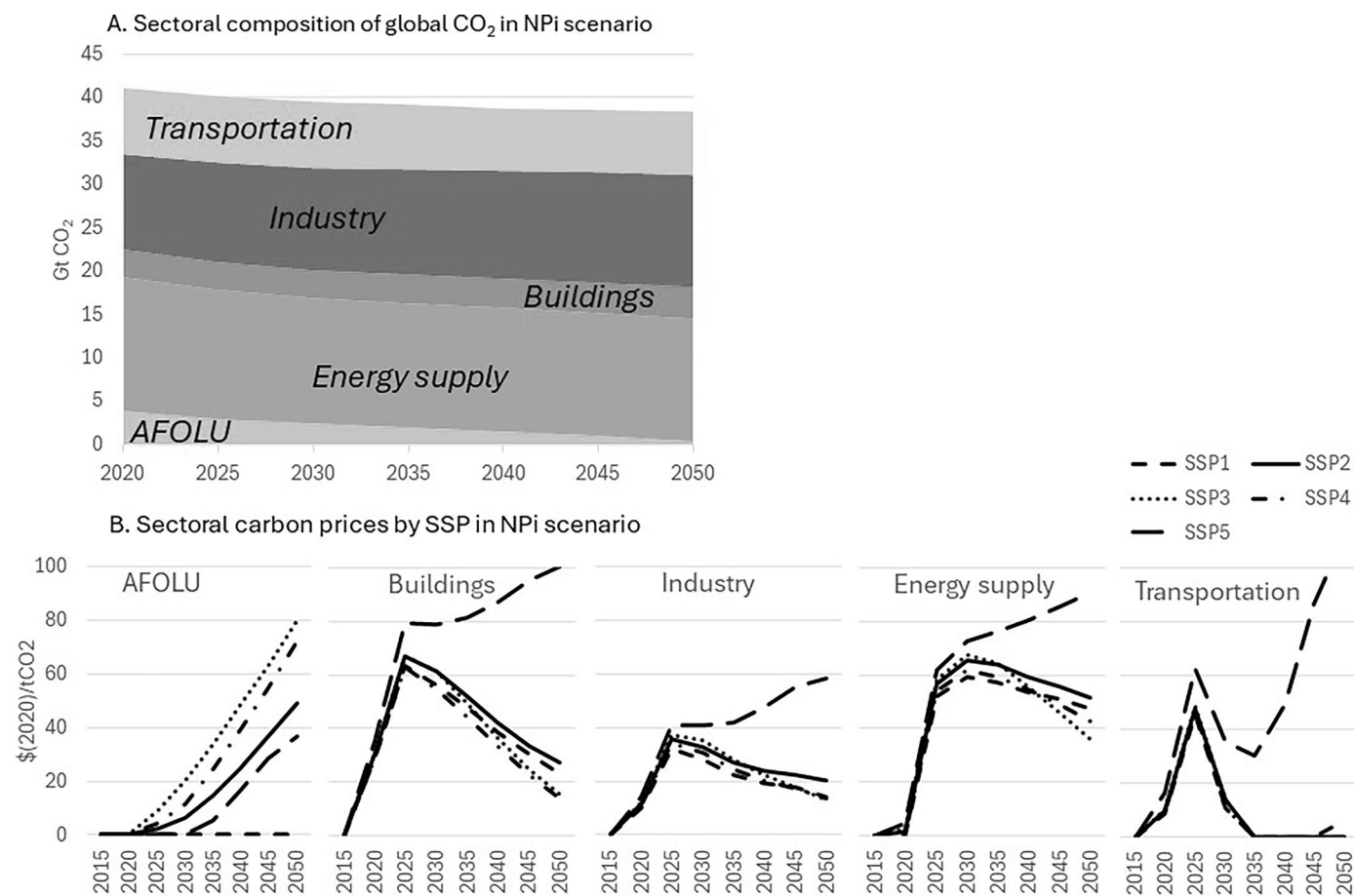
GCAM land use type	Matching land use types and intensity
Forest	Managed forests - Selective logging
Pasture	Pasture - Intense use
Irrigated crop	Cropland - Intense use
Rainfed crop	Cropland - Minimal use

Land use type mapping between land use type projections by GCAM and PSL CFs of specific land use types from Chaudhary & Brooks³¹. GCAM land uses not listed are considered natural vegetation.

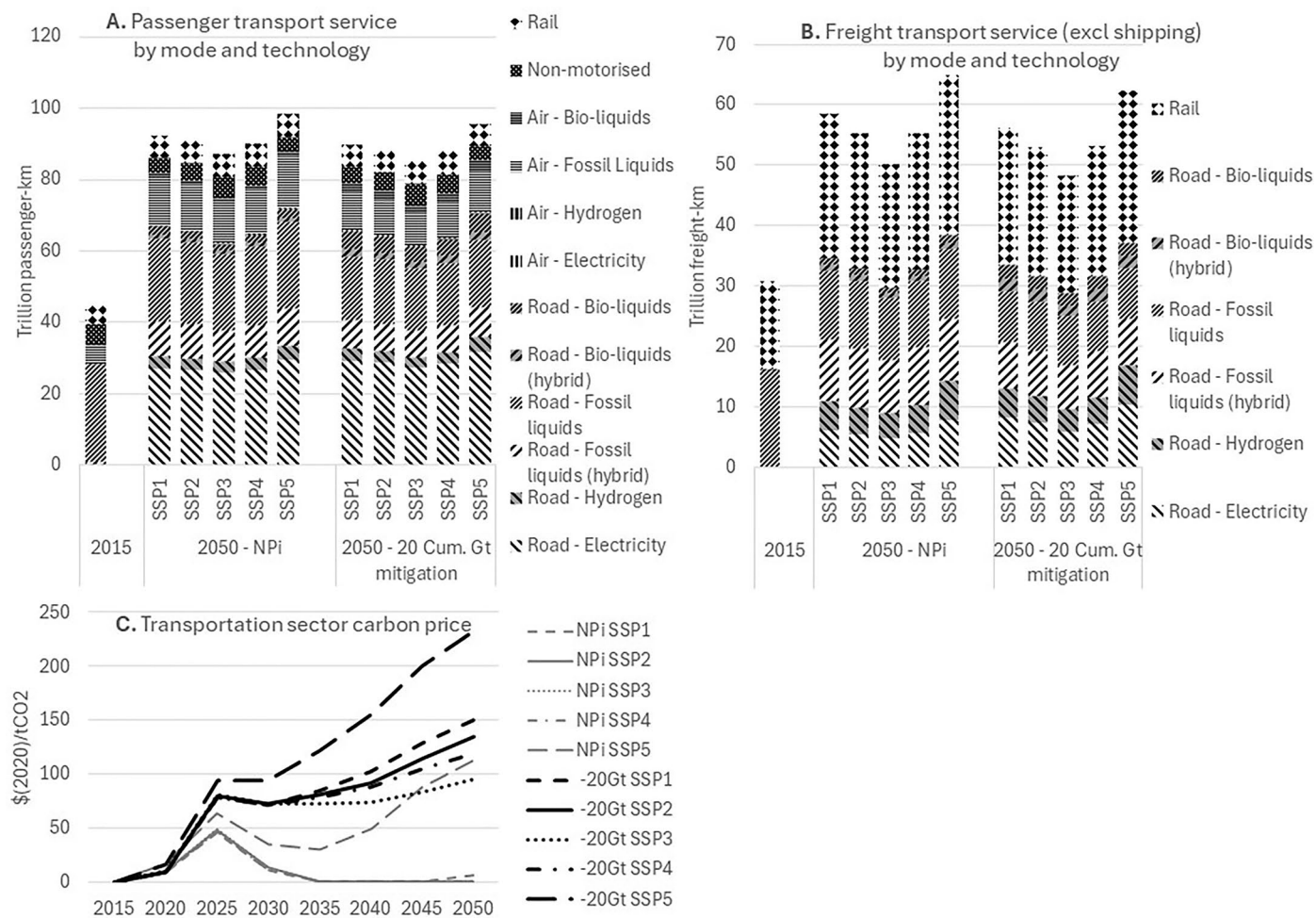


Extended Data Fig. 1 | Stocktake of sectoral emission and mitigation potentials. Sectoral emission stocktake for 2010, 2030 and 2050 for NPi baselines, 2 °C and 1.5 °C scenarios from IPCC AR6 (n values reflect number of models reporting, taking mean of all relevant scenarios from the same model; ranges refer to interquartile ranges), observed values in 2010 and 2020 (taken as average between 2019 and 2021 emissions to avoid Covid-19 distortions;

no historical value for AFOLU due to uncertainty), and applied “minimal benchmark” in this study. The 2050 point in the minimal benchmark is calculated as the lower 10% point in the total emissions range in 1.5 °C compatible scenarios for each sector, while the 2030 value in this benchmark is calculated adapting the path to the 2050 level departing from observed 2020 emissions.

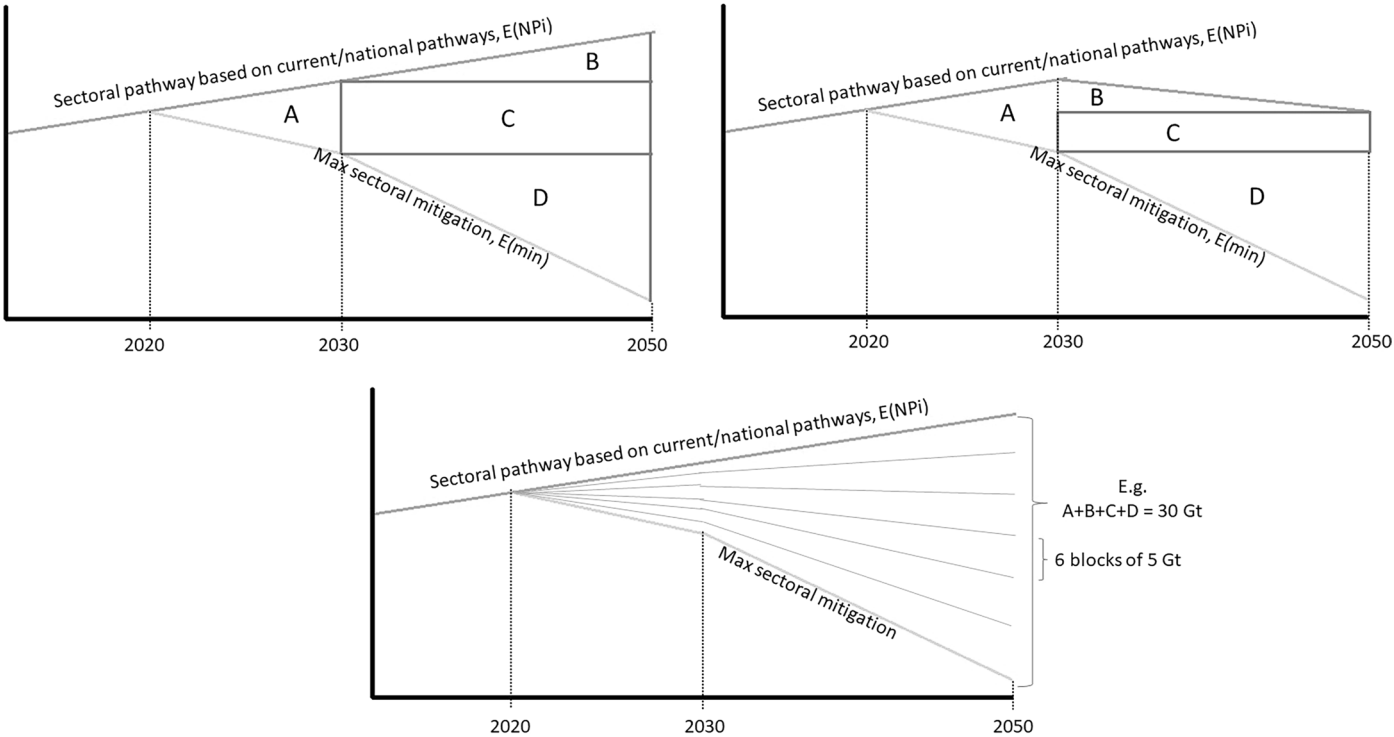


Extended Data Fig. 2 | NPi reference scenarios by SSP. **A.** Sectoral CO₂ emissions (for SSP2, nearly identical in other SSPs as determined by defined emission constraints based on stocktake analysis visualised in Table 1 and Extended Data Fig. 1). **B.** Carbon prices by sector and SSP.



Extended Data Fig. 3 | Graphical example of mitigation within transportation sector. **A.** Graphical example of how mitigation targets in the transportation sector (for the NPi level, see Table 1, and a 20 Gt cumulative CO₂ reduction, equal to the 1.5C-compatible SDG-balanced portfolio reflected in Fig. 5) translate to

a mix of mitigation options (electrification, biofuels, mode switch, demand reduction) for passenger transport. **B.** Idem for freight transport (excluding freight shipping for illustrative purposes). **C.** Heterogenous carbon prices, depending on the SSP narrative applied.



Extended Data Fig. 4 | Measuring cumulative mitigation blocks. Graphical sketch of how emission budgets by sector are calculated and split in blocks.